Adding and Subtracting in Scientific Notation

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lesson 1 — Adding and Subtracting in Scientific Notation	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.5

2. Standards

MA.8.NSO.1.5 Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

Must. Add and subtract numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

Must not. The two exponents stay within 2 of each other — that is this benchmark's own limit. Multiplication and division are 4.02–03; significant digits are 4.04.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — the powers of ten are set against the place-value names, so 'the powers must match' is read off the structure rather than announced as a rule.
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Boards 3 and 4 — the coefficient is checked against $1 \le a < 10$ before the board goes up, in both directions.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 6 — when the room splits between 1.23×10^7 and 12.3×10^6, which have the same value, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can add and subtract two numbers written in scientific notation, and write the answer in scientific notation.

Essential question. Why must two numbers written in scientific notation carry the same power of ten before they can be added?

Building on. MA.8.NSO.1.4 (3.08–3.09) — scientific notation, the coefficient rule and reading notation back; Grade 6 — adding and subtracting decimals by place value.

Working toward. 4.02–03 — multiplying and dividing in scientific notation; MA.8.NSO.1.6 (4.04) — real-world problems and significant digits.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	12	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why can 5.2×10^6 and 3.1×10^6 be added straight away, while 5.2×10^6 and 3.1×10^5 cannot?	DOK 2
Devon adds the exponents and gets 10^6. What would have to be true about the expression for adding exponents to be the right move?	DOK 3
12.3×10^6 and 1.23×10^7 are the same number. Why does one of them count as an answer and the other does not?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$(3.2 \times 10^5) + (4.1 \times 10^5)$	***7.3 × 10⁵***	7.3×10^{10} — the exponents were added as well as the coefficients [B1G-M 8.NSO.1.5 misconception: exponents added]; 1.312×10^{11} — the numbers were multiplied [8.NSO.1.5: the operation is addition]
2	A stadium sold 4.2×10^4 tickets on Friday and 3.5×10^4 on Saturday. Each ticket cost 38 dollars. How many tickets were sold on the two days?	***7.7 × 10⁴***	anything built from 38 — the unneeded figure was used [the question asks for a count, not a cost]; 7.7×10^8 — the exponents were added [B1G-M 8.NSO.1.5 misconception]
3	$(8 \times 10^{-6}) + (3 \times 10^{-6})$	***1.1 × 10⁻⁵***	11×10^{-6} — right value, but 11 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^9 instead of 1.2×10^{10}]; 1.1×10^{-11} — the exponents were added, giving 11×10^{-12}, and the answer was then finished
4	$(5.6 \times 10^8) - (4.9 \times 10^8)$	***7 × 10⁷***	0.7×10^8 — right value, but 0.7 is not at least 1 [8.NSO.1.5: the coefficient satisfies $1 \le a < 10$]; 1.05×10^9 — added instead of subtracting
5	$(1.6 \times 10^4) + (2.3 \times 10^6)$. Which is the sum?	***C — 2.316 × 10⁶***	(A) the coefficients were added without matching the powers of ten first [8.NSO.1.5: the powers must match]; (B) the coefficients were added and the exponents were added [B1G-M 8.NSO.1.5 named misconception: $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$]; (D) 1.6×10^4 was rewritten as 0.16×10^6 — the decimal point moved one place too few [8.NSO.1.5: the point moves as many places as the exponent changes]

#	What the board asks	Answer	If the board is wrong
6	$(2.5 \times 10^6) + (9.8 \times 10^6)$. Which is the sum, written in scientific notation?	***D — 1.23 × 10⁷***	(A) equal in value, but 12.3 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^9 instead of 1.2×10^{10}]; (B) the exponents were added as well as the coefficients [8.NSO.1.5: the power of ten does not change]; (C) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]
7	$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$	***7.16 × 10⁻²***	1.4×10^{-2} — the coefficients were subtracted without matching the powers [8.NSO.1.5]; 7.16×10^{-5} — the exponents were added, −2 + (−3) [8.NSO.1.5: the power of ten is a label, not a term]
8	$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$. What is the value of n?	***n = 5.7***	−3.48 — the target was not rewritten over 10^6 before subtracting [8.NSO.1.5: match the powers first]; 5.7×10^6 — the blank holds a coefficient, not a whole term
9	Devon writes $(1.9 \times 10^2) + (6.6 \times 10^4) = 8.5 \times 10^6$. Name the two mistakes he made, and give the correct sum.	***6.619 × 10⁴***	'he added wrong' — no mistake is actually named [the item asks which two]; 8.5×10^4 — the exponent mistake was caught and the matching mistake was not

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the place-value strip from Notes I with the power written above each name, and let the student write both numbers as words first — 'four point nine five ten-thousands' and 'twelve point one ten-thousands' — before any symbol is written. The matching step becomes reading, not procedure.

ELL. Coefficient and power of ten are the two phrases every question uses. Write 7.7×10^4 once with COEFFICIENT under the 7.7 and POWER OF TEN under the 10^4, and point at the labels on every board. Add or subtract is the third phrase: read each board's operation aloud before the boards go up.

Enrichment. Ask for $(4.2 \times 10^6) + (3.9 \times 10^5) + (1.1 \times 10^4)$ — three terms, all within two of each other — and then why the benchmark stops at a gap of two.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to name both of Devon's mistakes separately and give 6.619×10^4. Read the boards, not the papers.

Vocabulary. **scientific notation**; **coefficient**; **common power of ten**

Materials. Whiteboards and markers. Calculators are allowed on everything; entering 8E−6 + 3E−6 and reading 1.1E−5 back off the display is worth thirty seconds at whiteboard 3.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Add and subtract numbers written in scientific notation — HUR.